

Batch: 13 Roll No.: 16010324044
Experiment / assignment / tutorial No. 2
Grade: AA / AB / BB / BC / CC / CD / DD
Signature of the Staff In-charge with date

TITLE: Clipper circuit.

AIM: To analyse given clipper circuit. To design clipper circuit for given transfer characteristics.

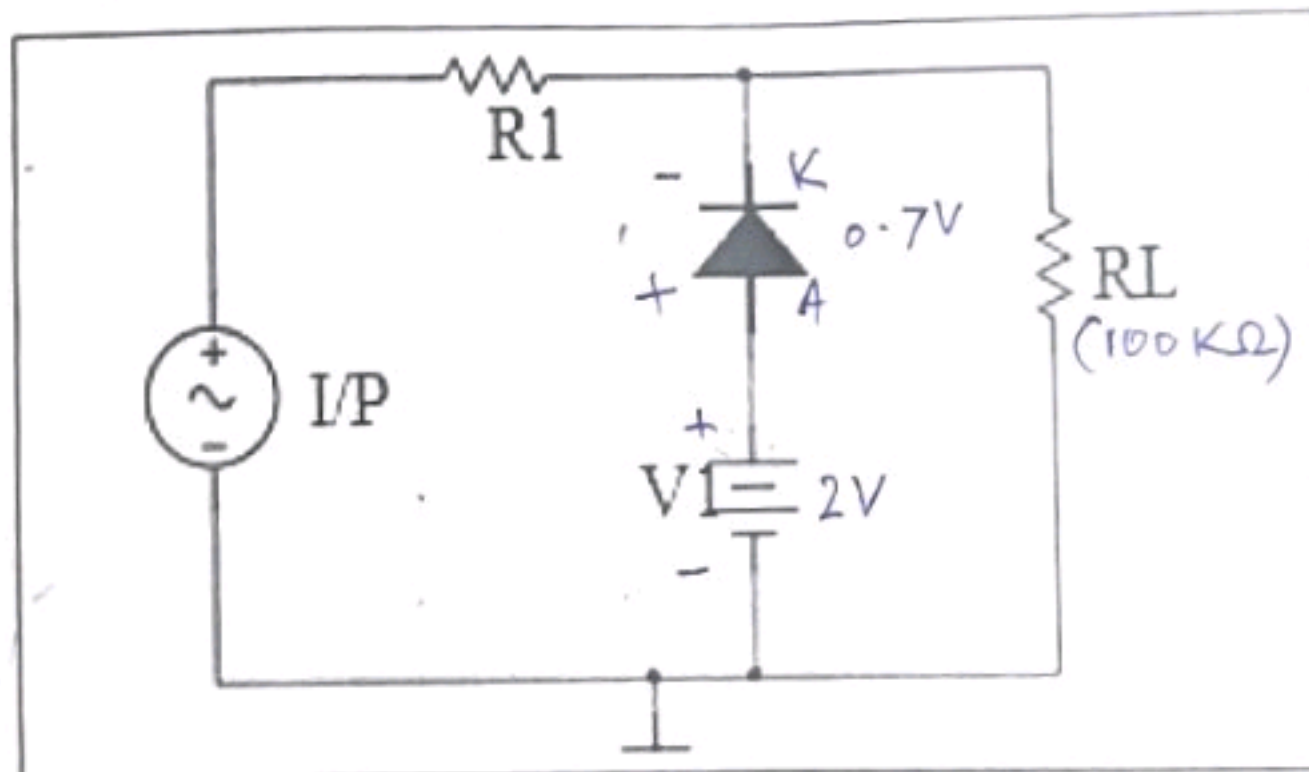
1. **OUTCOME:** Student will be able to analyze and design diode circuits

Part I- Analysis of clipper circuit

Procedure:

1. Develop the circuit in hardware as shown in the diagram.
2. Apply sinusoidal waveform 10V (P-P) as input and DC voltage (V1) at the given value of 2 V. Select $R1 = 1\text{ K}$ and $R_L = 100\text{ K}$.
3. Observe the waveforms.
4. Identify the type of Clipper circuit.
5. Draw voltage transfer characteristics.
6. Increase the DC voltage (V1) to 4V, and notice to what voltage are the positive peaks chopped off.
7. If the DC source is reversed, how does this affect the clipping?

Circuit Diagram:



→ Shunt Clipper

when,

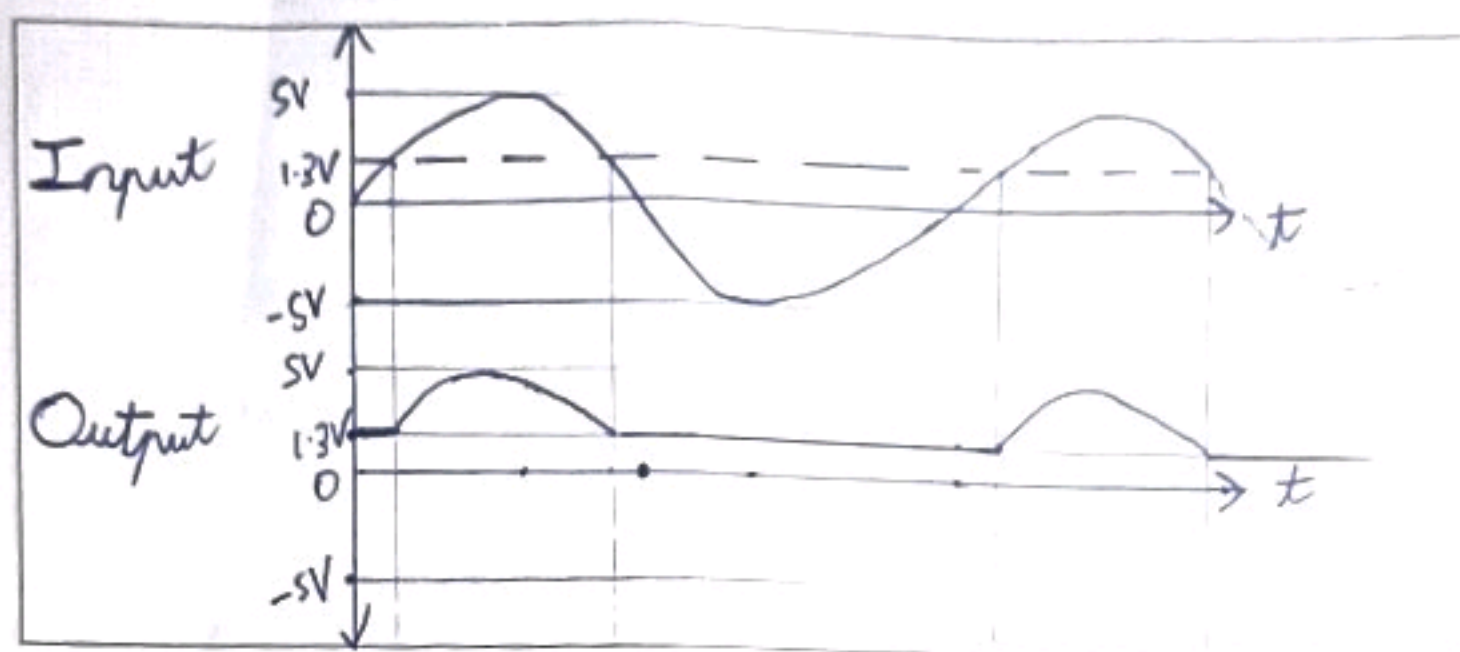
- $V_i < 1.3\text{V}$, Diode: FB, $V_o = 1.3\text{V}$
- $V_i > 1.3\text{V}$, Diode: RB, $V_o = V_i$

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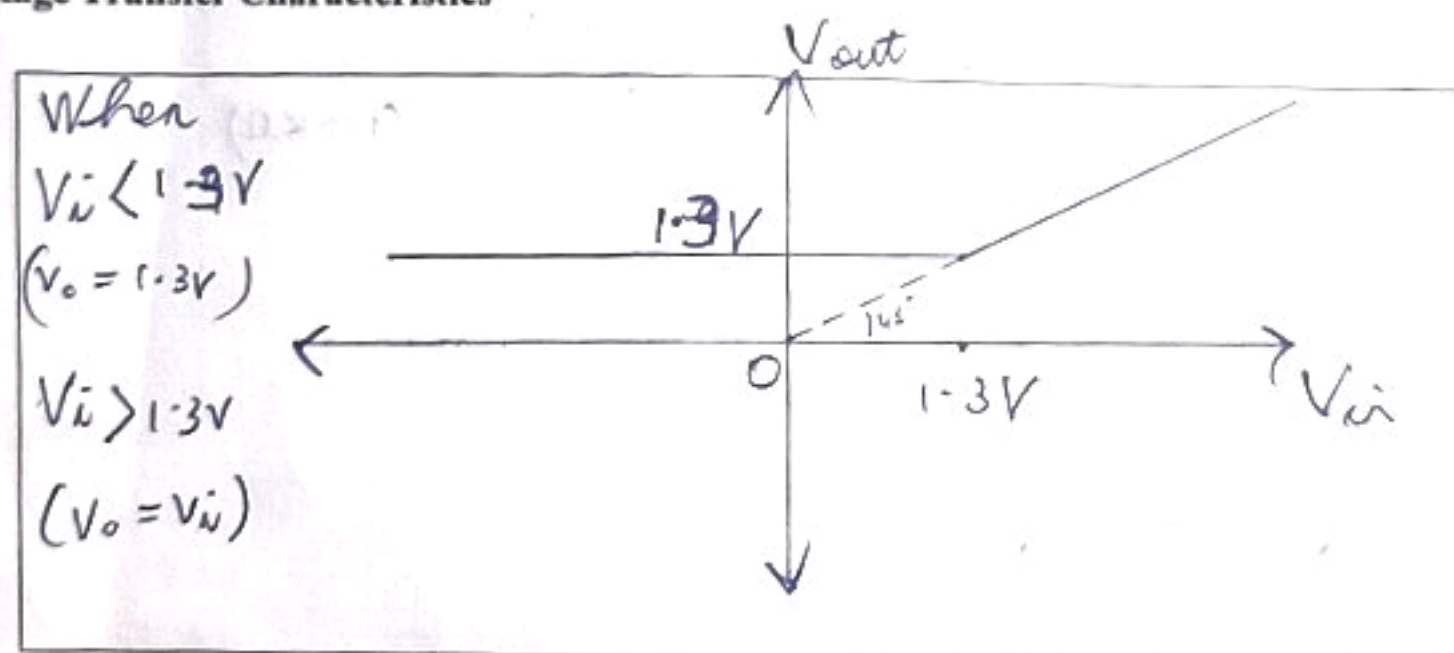
Observation Table I:

	V_{in} (p-p)	V_{out} (p-p)	Biassing Voltage (Clipping levels)
Theoretical	10	3.7V	1.3V
Practical	9.4	3.4V	1.13V

Input and output waveforms

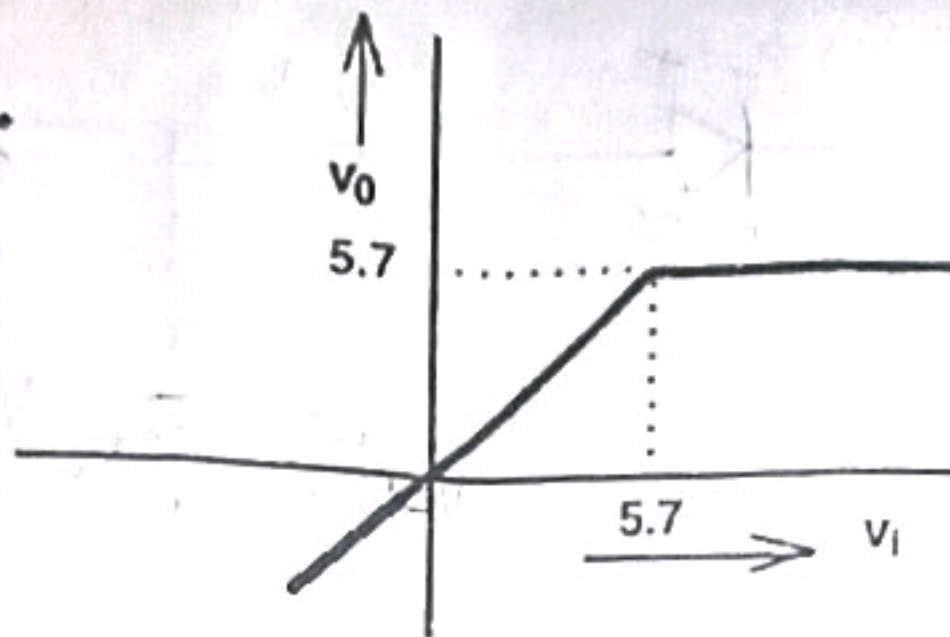


Voltage Transfer Characteristics



Part II- Design of Series Clipper

Design a series clipper that will yield the voltage transfer function shown in figure.
Assume diode cut-in voltage of $V_\gamma = 0.7V$

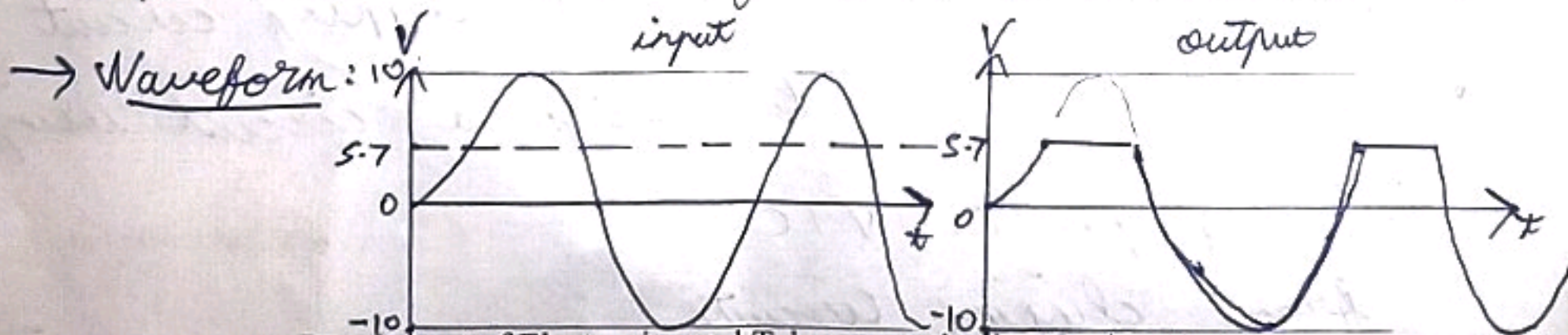


Procedure:

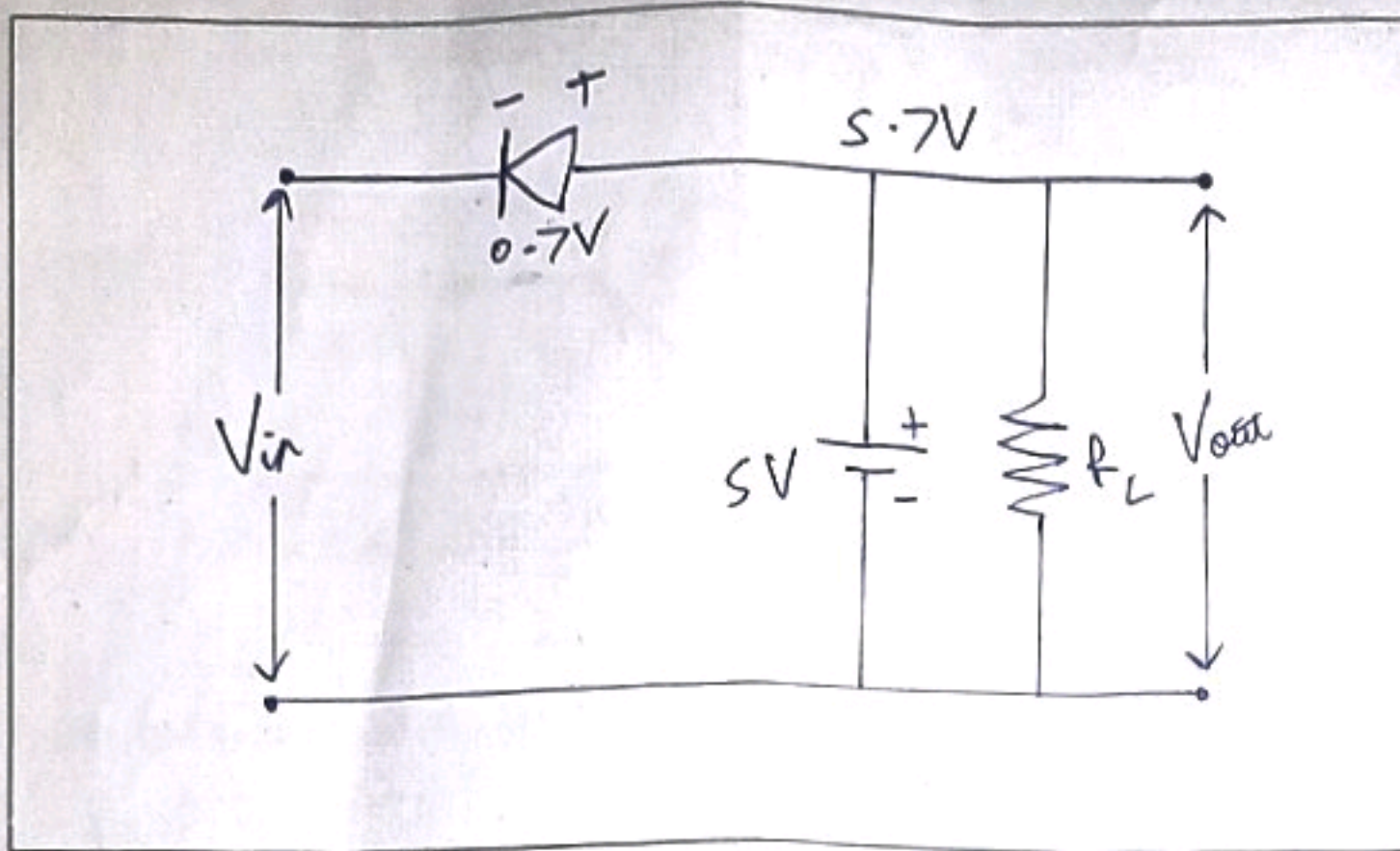
1. Use sinusoidal waveform as input.
2. Set the clipping levels (biasing voltage) as per the voltage transfer function.
3. Implement the designed circuit in hardware
4. Plot input, output waveforms.

Design steps:

- • The graph shows the v_d ~~but~~ b/w V_i (i/p voltage) & V_o (o/p voltage)
- For $V_i < 5.7$, ($V_o = V_i$), this indicates that for the input voltages below clipping levels, the ckt. acts as a pass through, & o/p follows i/p linearly.
 - For $V_i > 5.7$, (V_o) is clipped at $5.7V$, this signifies that the diode becomes F.B. and conducts, effectively limiting the o/p voltage to ~~the~~ bias voltage ($5.7V$ in this case).



Circuit Diagram



Observation Table II:

	V_{in} (p-p)	V_{out} (p-p)	Biassing Voltage (Clipping levels) -
Theoretical	19.8	13.9	5.8
Practical	20	14.3	5.7

Conclusion:

- We learnt about various clipping circuit
- We learnt how to design a circuit using VTC graph & VTC from clipping circuit

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